ATOMIC STRUCTURE: MASTER TEXTBOOK

1. Development of Atomic Models

From Dalton to Bohr

- **Rutherford's Model**Proposed the nuclear model with electrons orbiting a tiny, positively charged nucleus.
- **Bohr's Model:**Suggested that electrons revolve in discrete, fixed-energy circular orbits (stationary states).

2. -Quantum Mechanical Model

Dual Nature of Matter

De Broglie's hypothesis: Matter (like electrons) has both wave-like and particle-like properties. Heisenberg's Uncertainty Principle states it is impossible to simultaneously determine the precise position and momentum of a subatomic particle.

3. Quantum Numbers

Defining Electron State

Four quantum numbers uniquely define the state of an electron in an atom:

- **Principal (n):** Size and energy level.
- **Azimuthal (l):** Shape of the orbital (s, p, d, f).
- **Magnetic (m_l):** Orientation in space.
- **Spin (m_s):** Spin orientation of the electron (+1/2, -1/2).

4. Electronic Configuration Rules

Filling Orbitals

- **Aufbau Principle:** Orbitals are filled in increasing order of energy.
- **Pauli Exclusion Principle:** No two electrons can have the same set of all four quantum numbers.
- **Hund's Rule:** Pairing in degenerate orbitals happens only after all are singly occupied.